Lab #1: Introduction to Laboratory Electronics

Name: Tony Han

Partners at Table: Pranay, Arta, Marnu

1. Experiment Description and Prelab

1.1: Description

(Give a brief description of the experiment and objectives for context.)

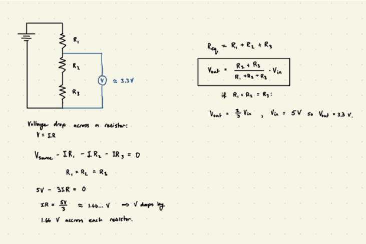
In this experiment, we are to measure 2 waveforms from an Arbitrary Waveform Generator (AWG): a 50kHz sinewave, and a 200kHz square wave. Then, we have to verify that the circuit in the prelab (seen later in this document) actually behaves as we expect it to. With this section, we can experiment to find out the significance between the scope(+) and scope(-) leads. Afterwards, we will build our actual voltage divider according to the prelab design. Here, we will use three 10 kilo-Ohm resistors, and measure our actual voltage output. Account for error and error propagation! We then should find the amount of power we lose by building the circuit as such, and look for changes to reduce the amount of power lost. Then, we must find out how to change our circuit to actually function as a 3.3V supply, and then build the re-design whilst confirming it works with recorded observations.

1.2 Prelab: Voltage Divider

(Give problem statement, insert design of voltage divider and derive expression of Vout as function of R1,R2,R3 and Vin. Include circuit diagrams!)

In the prelab, we are meant to design a voltage divider to reduce a 5V source to supply 3.3V, using 3 identical resistors. My design of the circuit, with all associated equations are as shown in the image below:

In [ ]:
```
InlineImage('IMG_1113.png', width=8, height=8, caption= "Figure 1, prelab circuit design", fs=24);
```



$R_{eq} = R_1 + R_2 + R_3$

$$\boxed{V_{out} = \frac{R_2 + R_3}{R_1 + R_2 + R_3} \cdot V_{in}}$$

if $R_1 = R_2 = R_3$:

$V_{out} = \frac{2}{3} V_{in}$, $V_{in} = 5V$ so $V_{out} = 3.3$ V.

Voltage drop across a resistor:
$V = IR$

$V_{source} - IR_1 - IR_2 - IR_3 = 0$

$R_1 = R_2 = R_3$

$5V - 3IR = 0$

$IR = \frac{5V}{3} \approx 1.66\ldots$ V → V drops by
1.66 V accross each resistor.

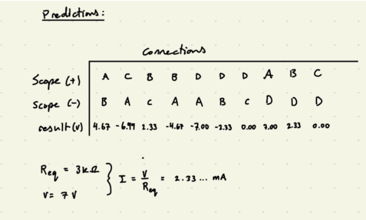
Figure 1, prelab circuit design

1.3 Prelab: AC/DC/GND

(Copy the table from the prelab, include the circuit diagram and fill it in with your reasoning)

In [4]:
```
InlineImage('IMG_1116.png', width=8, height=8, caption= "Figure 2, prelab circuit predictions", fs=24);

InlineImage('circuitsample.png', width=8, height=8, caption= "Figure 1, prelab circuit design", fs=24);
```

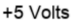


Predictions:

	Connections									
Scope (+)	A	C	B	B	D	D	D	A	B	C
Scope (–)	B	A	C	A	A	B	C	D	D	D
result (v)	4.67	-6.99	2.33	-4.67	-7.00	-2.33	0.00	7.00	2.33	0.00

$\left.\begin{array}{l} R_{eq} = 3k\Omega \\ V = 7V \end{array}\right\} I = \frac{V}{R_{eq}} = 2.33\ldots$ mA

Figure 2, prelab circuit predictions



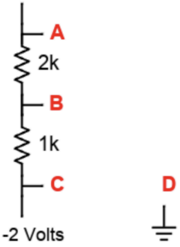


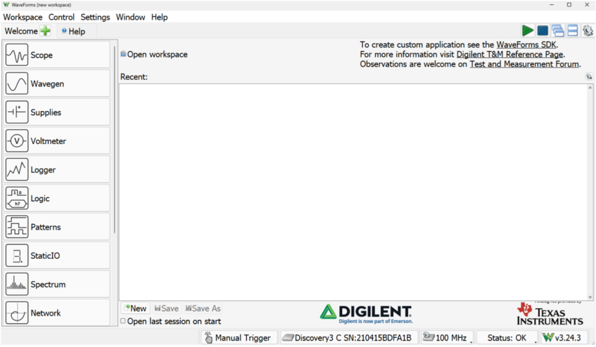
Figure 1: Simple Resistor Circuit.

Figure 1, prelab circuit design

Essentially, my predictions were off calculating the current through the circuit, and then using it to calculate the voltage drop accross any set or individual resistor. However, some parts were confusing; such as the drop from B to D being the same as C to D in my predictions, even though logically C is -2V and D is ground or 0V, so there should be a difference between those two scenarios...? Maybe actually building it tomorrow will provide insight as to why.

1.4 Prelab: Analog Discovery 2

```
In [7]: InlineImage('waveforms.png', width=13, height=8, caption= "Figure 3, screenshot of connected AD3 in waveforms.", fs=24);
```



To create custom application see the WaveForms SDK.
For more information visit Digilent T&M Reference Page.
Observations are welcome on Test and Measurement Forum.

Figure 3, screenshot of connected AD3 in waveforms.

2. Oscilloscope and AWG

2.1: Task #1: 50 KHz SineWave

(Describe the steps you took to setup your AD2 and the connections you made with the pins. Sketch this setup.)

To set up my AD3, I followed the guide on the AD3 website - feedback loop circuit, where I connect: orange 1+ to yellow W1, and orange 1- to gnd. Pin connections shown below:

```
In [10]: InlineImage('pinsetup.png', width=8, height=8, caption= "Figure 4, AD3 connections", fs=24);
```

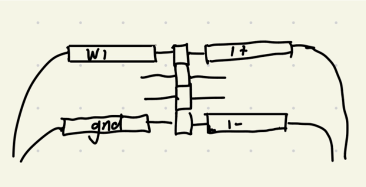
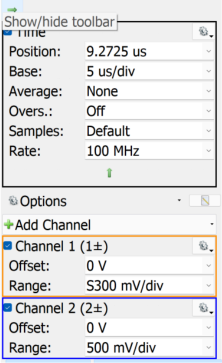


Figure 4, AD3 connections

(Describe how you took measurements and their results)

Then, I set up a 50kHz sine wave, using the following window settings:

```
In [12]: InlineImage('sinesettings.png', width=8, height=8, caption= "Fig. 5, scope settings", fs=24);
```



Time
Position: 9.2725 us
Base: 5 us/div
Average: None
Overs.: Off
Samples: Default
Rate: 100 MHz

Channel 1 (1±)
Offset: 0 V
Range: S300 mV/div

Channel 2 (2±)
Offset: 0 V
Range: 500 mV/div

Fig. 5, scope settings

The measurements of the peak-peak voltage and period with the cursor are shown below:

```
In [16]: InlineImage('amplitude.png', width=8, height=8, caption= "Figure 6, sine waveform voltage", fs=24);
```

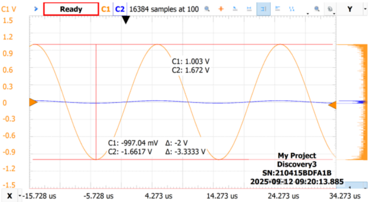


Figure 6, sine waveform voltage

```
In [14]: InlineImage('Period.png', width=8, height=8, caption= "Figure 7, sine waveform period", fs=24);
```

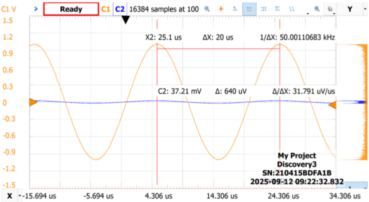


Figure 7, sine waveform period

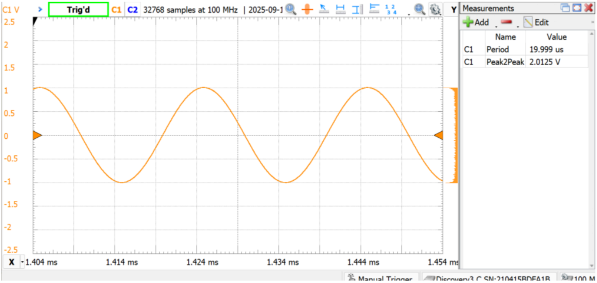
Peak to peak = 2v

Period = 20 us

Since this measurement can be skewed by human error, I will take uncertainty for this measurement based off of the next section; the quick measure tool.

Now using the quick measure tool:

```
In [18]: InlineImage('quicksine.png', width=13, height=8, caption= "Figure 8, sine waveform quick measurements", fs=24);
```



	Name	Value
C1	Period	19.999 us
C1	Peak2Peak	2.0125 V

Figure 8, sine waveform quick measurements

The period would fluctuate between 19 and 20; given that it is a digital measurement, I will use the half of the range of flucuation (the +/-) and divide by square root 3:

$$uncertainty = \frac{0.5}{sqrt(3)} = \pm 0.28867....us$$

However, due to the nature of uncertainty being, well... uncertain, I will round it to the first significant figure:

$$uncertainty_{period} = \pm 0.3us$$

The amplitude would fluctuate between 2.00 and 2.01; following the same logic as above, we obtain the uncertainty of:

$$uncertainty_{amplitude} = \pm 0.003V$$

So:

$$Period = 19.5 \pm 0.3us, and Amplitude = 2.005 \pm 0.003V.$$

These values (19.5 and 2.005) were chosen because they stayed on each value approx evently. So I picked the value in between them.

(Note any difficulties or troubles and how they were corrected)

No troubles in this step yet. I had previously tested the AD3 prior to lab time. Note: cursor feels awkward to use since it is placing 2 bars for each axis to measure.

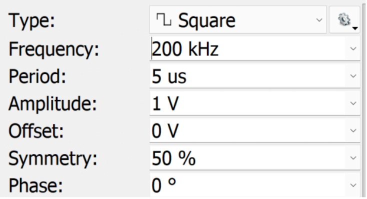
2.2 Task #2: 200 KHz Square Wave

(Describe you adjusted the AWG to produce desired output)

To produce the 200kHz square wave, I simply switched the wave type to square, and then set the frequency to 200kHz.

In [20]:

```
InlineImage('squaresettings.png', width=8, height=8, caption= "Figure 9, square wave settings", fs=24);
```



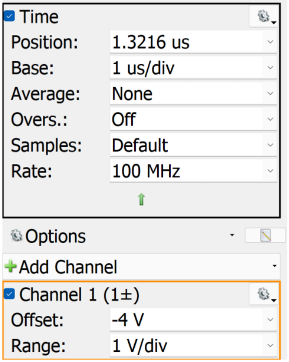
Type: Square
Frequency: 200 kHz
Period: 5 us
Amplitude: 1 V
Offset: 0 V
Symmetry: 50 %
Phase: 0 °

Figure 9, square wave settings

However, I am having troubles with the scope settings to see the wave. It seems as if the wave won't show up no matter the range or base settings. Turns out, there was an x offset that I had from using the scroll to zoom in/out. I can see the wave upon resetting this offset. Updated scope settings:

In [21]:

```
InlineImage('squarescope.png', width=8, height=8, caption= "Figure 10, good scope settings", fs=24);
```



☑ Time

Position: 1.3216 us

Base: 1 us/div

Average: None

Overs.: Off

Samples: Default

Rate: 100 MHz

Options

Add Channel

☑ Channel 1 (1±)

Offset: -4 V

Range: 1 V/div

Figure 10, good scope settings

```
In [26]: InlineImage('cursor2.png', width=8, height=8, caption= "Figure 11, square waveform Period", fs=24);
```

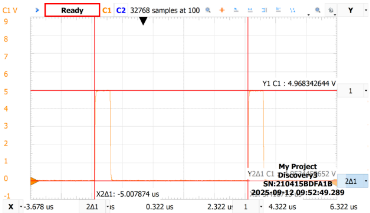


Figure 11, square waveform Period

measurements of square wave

I realized at this point in the lab that the embedded images does not work, so I have to split the cell and insert a new code section for images; images before this point will not show up unfortunately, and I will have to reformat in my final copy.

NOTE: I will also determine uncertainties at home for the sake of time, since we only have access to resistors in lab. The waves I can recreate according to my notes at home with ease.

Below is my graph from the exported data for the square wave: NOTE: in lab, I was not able to get the data to import correctly. No matter what I and the TAs tried, the file directory could not be found. Will look into it further at home, or maybe find an alternative file/image import method. We tried every directory possible, including the local and non-local directories, even re-exporting/importing data, and also refreshed the syzygy instance. Nothing worked. For this submission, images may not be present for that reason.

POST LAB TONY HERE, I have gone through a fresh installation of python, jupyter, and all other softwares needed on my computer to use jupyter notebook through VSCc

I realized at this point in the lab that the embedded images does not work, so I have to split the cell and insert a new code section for images; images before this point will not show up unfortunately, and I will have to reformat in my final copy.

NOTE: I will also determine uncertainties at home for the sake of time, since we only have access to resistors in lab. The waves I can recreate according to my notes at home with ease.

Below is my graph from the exported data for the square wave:

NOTE: in lab, I was not able to get the data to import correctly. No matter what I and the TAs tried, the file directory could not be found. Will look into it further at home, or maybe find an alternative file/image import method. We tried every directory possible, including the local and non-local directories, even re-exporting/importing data, and also refreshed the syzygy instance. Nothing worked. For this submission, images may not be present for that reason.

POST LAB TONY HERE: I have gone through a full reinstallation of everything needed to run jupyter notebook locally through visual studio code.
Now, I am able to upload the images properly as the computer seems to have no issues finding file directories now. Sadly I had to manually re-upload each one :*(

```
In [41]: #x = np.linspace(0,2*np.pi,50) # Creates an array of 50 equally spaced floats between 0 and 2pi
         #sinewave = np.sin(x) # Computes the sine of each element in 'x' and stores the result in 'sinewave'

         fig,ax = plt.subplots(figsize=(8,8)) # Generates a figure of size 8x8 along with an axis object

         data = pd.read_csv("squarewavedata.csv")
         t = data['Time (s)'] #Columns in pandas dataframes can be indexed by their header
         V = data['Channel 1 (V)']

         ax.plot(t, V, color='r', linewidth=2, zorder=1, label='Line plot') # Line plot 'sinewave' vs. 'x' in red with 2 point linewidth
         ax.scatter(t, V, color='k',s=2, zorder=2, label='Scatter plot') # Scatter plot 'sinewave' vs. 'x' in black with 20pt dots. The zorder argument ensures the s
         ax.set_xlim(0,0.000015)
         # #ax.set_xticks([0,np.pi/2,np.pi,3*np.pi/2,2*np.pi]) # Matplotlib will automatically generate ticks, but you can set them manually by passing an array like
         # ax.set_xticklabels(['0','$\pi/2$','$\pi$','$3\pi/2$','$2\pi$']) # Rather than display fractions of pi as decimal values, we can replace the tick labels wi

         # Make sure your fonts are large enough to be legible!
         ax.tick_params(axis='x', labelsize=14)
         ax.tick_params(axis='y', labelsize=14)
```

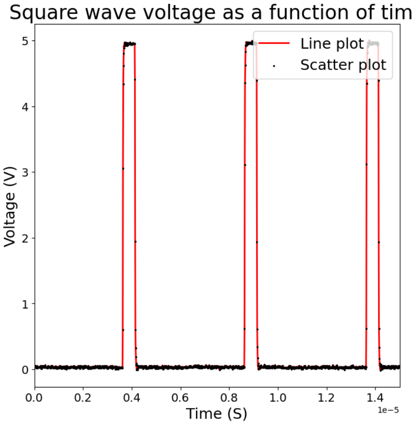
```
# Make sure to label your axes!
ax.set_ylabel('Voltage (V)',fontsize=18)
ax.set_xlabel('Time (S)',fontsize=18)

ax.set_title('Square wave voltage as a function of time', fontsize=24)

ax.legend(loc='upper right', fontsize=18) # We can include a simple legend like this - the labels assigned to each plot as an argument in their function cal
```

Out[41]:

```
<matplotlib.legend.Legend at 0x19ff509ead0>
```



(Note any difficulties or troubles and how they were corrected)

Initially, the graph showed too much data; looked like a bunch of vertical lines. I had to limit the range of x axis data, so that it only shows 3 square waves so it is clearer.

2.3 Task #3: Differential vs GND references

(Build the circuit from the prelab Fig. 1. Include diagram and pic.)

Pictures will be uploaded later once the problem of uploading is resolved.

For now, I will describe what I have completed with the AD3. I have created a new supply, setting the positive terminal to +5v and the negative to -2V. After that, I can set up the breadboard corresponding to how the image is. The probes for the oscilloscope (1+ and 1-) were shifted to positions A,B,C,D accordingly to obtain measurements. The measurement average tool was used to obtain an average of the voltage reading between the 1+ and 1- scope probes.

POST LAB TONY HERE: The circuit diagram is now shown below, with a picture of it:

In [45]:

```
InlineImage('circuitpic.jpg', width=8, height=8, caption= "Figure 12, breadboard setup", fs=24);
InlineImage('circuitdiagram.png', width=8, height=8, caption= "Figure 13, circuit diagram", fs=24);
InlineImage('power.png', width=13, height=8, caption= "Figure 14, power supply settings", fs=24);
```

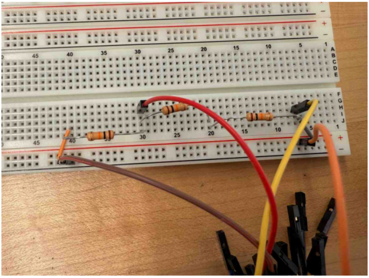


Figure 12, breadboard setup

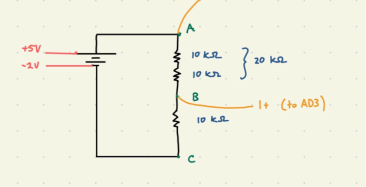
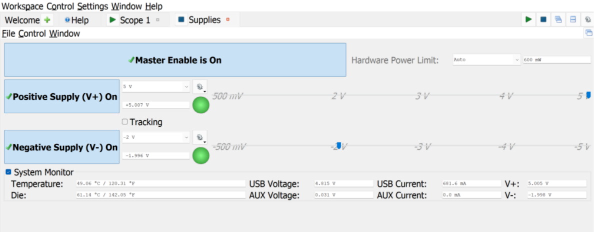


Figure 13, circuit diagram



✓Master Enable is On

Hardware Power Limit: Auto 600 mW

✓Positive Supply (V+) On 5 V +5.007 V

☐ Tracking

✓Negative Supply (V-) On -2 V -1.996 V

☑ System Monitor

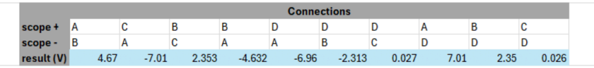
Temperature: 49.06 °C / 120.31 °F
Die: 61.14 °C / 142.05 °F
USB Voltage: 4.815 V
AUX Voltage: 0.031 V
USB Current: 681.6 mA
AUX Current: 0.0 mA
V+: 5.005 V
V-: -1.998 V

Figure 14, power supply settings

Initially, I had the position of the 1+ and 1- probes reversed, and my window settings did not allow me to see the waveform. However, upon switching them around, I found out what was wrong. The reading was in the negative axis but my scope was in the positive axis. (because the probes were reversed)

My results from the actual measurements of the circuit are shown below:

```
In [53]: InlineImage('resultsl.png', width=13, height=8, caption= "Figure 15, measurement results", fs=24);
         InlineImage('uncert.png', width=13, height=8, caption= "Figure 16, measurement fluctuation", fs=24);
```



	Connections									
scope +	A	C	B	B	D	D	D	A	B	C
scope -	B	A	C	A	A	B	C	D	D	D
result (V)	4.67	-7.01	2.353	-4.632	-6.96	-2.313	0.027	7.01	2.35	0.026

Figure 15, measurement results

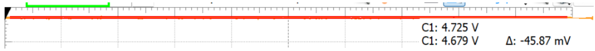


Figure 16, measurement fluctuation

(Note any difficulties or troubles and how they were corrected)

no difficulties in this stage. Had to be diligent in knowing which wire is which for ease of measurement. Negative signs were lost, so had to re-measure occasionally. Otherwise, procedurally everything was ok. SIDE NOTE: I have reported the result to the highest fluctuating digit. However, if I were to perform calculation on these values, I would use the range of the measurement fluctuation shown in figure 16, and divide it by square root 3 since it is a digital measurement.

(Answer red conceptual questions)

The difference between the positive and negative leads is that the positive one expects to be at a higher potential than the negative lead. This is presumably due to the measurement method, where voltage is relative to each lead. As such, by sign convention, the positive lead should be at a higher voltage point for the reading to be positive. However, if the leads are reversed and the negative lead is placed at a place where relative voltage is higher than the positive lead, the reading for the voltage drop will be negative. Essentially, it is just to provide a solid reference point to be able to measure the voltage drop between any two points in the circuit, without needing a ground (0 Volt) reference - the voltage at each point is relative to each other.

My prelab answers were quite close to the actual results. I remembered to switch the negative signs when the positive probe was placed furhter along the circuit than the negative probe. This is partly because in highschool, my physics teacher did a demo very similar to this except with parallel plates; the sign would reverse depending on probe placement; As such, I was able to consider this while calculating my predictions.

3. DC Voltage Divider

3.1 Task #4: Unloaded Voltage Divider

(Build your design from prelab, include diagram and pic and how AD2 breakout board was configured.)

Diagram will be inserted once I have the image upload figured out.

```
In [51]: InlineImage('diagramm.png', width=13, height=8, caption= "Figure 17, circuit diagram", fs=24);
         InlineImage('circuitimag.jpg', width=13, height=8, caption= "Figure 18, breadboard circuit", fs=24);
         InlineImage('powerset.png', width=16, height=8, caption= "Figure 19, power settings", fs=24);
```

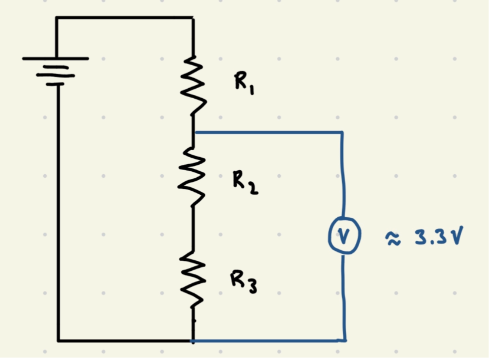


Figure 17, circuit diagram

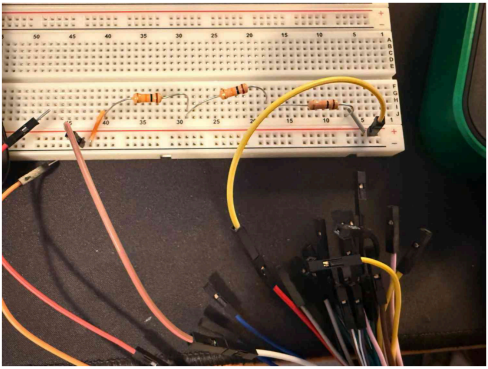
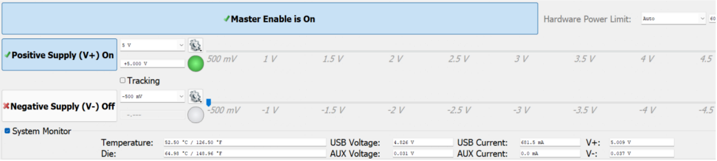


Figure 18, breadboard circuit



✓Master Enable is On

Hardware Power Limit: Auto

✓Positive Supply (V+) On 5 V +5.000 V

☐ Tracking

✗Negative Supply (V-) Off -500 mV -.---

☑ System Monitor

Temperature: 52.50 °C / 126.50 °F

Die: 64.98 °C / 148.96 °F

USB Voltage: 4.826 V

AUX Voltage: 0.031 V

USB Current: 681.5 mA

AUX Current: 0.0 mA

V+: 5.009 V

V-: 0.037 V

Figure 19, power settings

(Record measured Vout with uncertainty and compare to expected value with propagated uncertainty. Show/describe how you propagated expected error)

For the measured Vout:

$$V_{out} = 3.305V \pm 0.014V.$$

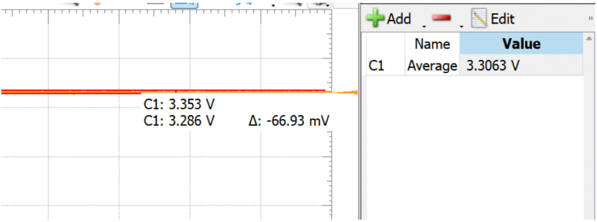
How was this obtained? Well, if we look at figure 20 below, we can see that there are two possible choices for uncertainty. One is the digital reading of the average, and the other is the actual fluctation of the signal, which I have measured to have a range of 66.93mV. Since the digital reading is unrealisticaly precise, given the amount of flucutation, I have chosen to use the uncertainty from the noise in the waveform. I divided the range by $2sqrt(6)$ since it is an analog measurement, and rounded to the largest non-1 or 0 digit (I was taught to do this before, I'm not sure if it applies here), resulting in an uncertainty of $\pm 0.014V$. I decided on 3.305 since the value fluctuated consistently between 3.31 and 3.30.

I think logically the value should be more like $V_{out} = 3.31V \pm 0.01V$. if uncertainty must be rounded to 1 significant figure (the rest is uncertain).

For the expected value of Vout (obtained by measuring each resistor and the voltage supply separately, then propagating the uncertainty for all measurements)

See image below! I have split up the process into different steps; Calculating each associated uncertainty, then the propogation in the first step of operations, then the propogation to the next step. Parts of the equation are colour coded for legibility.

```
In [65]: InlineImage('uncertain.png', width=13, height=8, caption= "Figure 20, measurement fluctuation", fs=24);
```

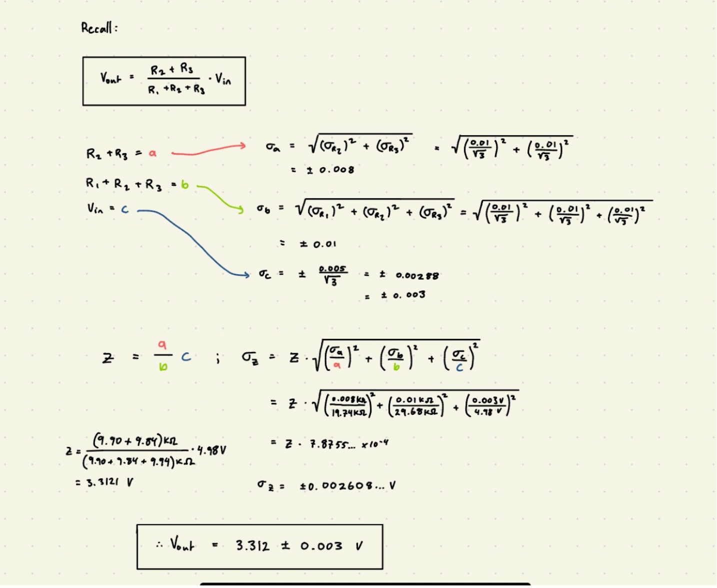


	Name	Value
C1	Average	3.3063 V

C1: 3.353 V
C1: 3.286 V
Δ: -66.93 mV

Figure 20, measurement fluctuation

```
In [64]: InlineImage('calculations.png', width=28, height=14, caption= "Figure 21, Uncertainty propogation", fs=24);
```



Recall:

$$\boxed{V_{out} = \frac{R_2 + R_3}{R_1 + R_2 + R_3} \cdot V_{in}}$$

$R_2 + R_3 = a$ $\longrightarrow$ $\sigma_a = \sqrt{(\sigma_{R_2})^2 + (\sigma_{R_3})^2} = \sqrt{\left(\frac{0.01}{\sqrt{3}}\right)^2 + \left(\frac{0.01}{\sqrt{3}}\right)^2}$

$= \pm 0.008$

$R_1 + R_2 + R_3 = b$ $\longrightarrow$ $\sigma_b = \sqrt{(\sigma_{R_1})^2 + (\sigma_{R_2})^2 + (\sigma_{R_3})^2} = \sqrt{\left(\frac{0.01}{\sqrt{3}}\right)^2 + \left(\frac{0.01}{\sqrt{3}}\right)^2 + \left(\frac{0.01}{\sqrt{3}}\right)^2}$

$= \pm 0.01$

$V_{in} = c$ $\longrightarrow$ $\sigma_c = \pm \frac{0.005}{\sqrt{3}} = \pm 0.00288$

$= \pm 0.003$

$$Z = \frac{a}{b} c \quad ; \quad \sigma_Z = Z \cdot \sqrt{\left(\frac{\sigma_a}{a}\right)^2 + \left(\frac{\sigma_b}{b}\right)^2 + \left(\frac{\sigma_c}{c}\right)^2}$$

$$= Z \cdot \sqrt{\left(\frac{0.008\,k\Omega}{19.74\,k\Omega}\right)^2 + \left(\frac{0.01\,k\Omega}{29.68\,k\Omega}\right)^2 + \left(\frac{0.003\,V}{4.98\,V}\right)^2}$$

$$Z = \frac{(9.90 + 9.84)\,k\Omega}{(9.90 + 9.84 + 9.94)\,k\Omega} \cdot 4.98\,V$$

$$= 3.3121\ V$$

$$= Z \cdot 7.8755\ldots \times 10^{-4}$$

$$\sigma_Z = \pm 0.002608\ldots\ V$$

$$\boxed{\therefore V_{out} = 3.312 \pm 0.003\ V}$$

Figure 21, Uncertainty propogation

***Results**

The two values, measured and calculated, are quite close. Neither value falls within the uncertainty range of the other, but if you account for the uncertainty of both values, then there is some overlap. This tiny difference could be due to several factors, including impurities in the air or dust taking away some electric charge while it is actually in the circuit. There could also be other factors such as the breadboard not being a perfect insulator (also removing minute quantities of charge?). This would make sense since the voltage out is lower in the actual circuit (by a tiny amount), so maybe some charge was lost, resulting in a slightly weaker current.

Below are just some images of my measurements.

```
In [63]: InlineImage('resistors.png', width=13, height=8, caption= "Figure 22, circuit resistor measured values", fs=24);
         InlineImage('voltage.png', width=13, height=8, caption= "Figure 23, ", fs=24);
         #InlineImage('powerset.png', width=16, height=8, caption= "Figure 19, power settings", fs=24);
```

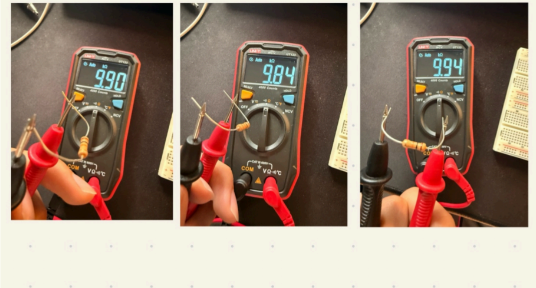


Figure 22, circuit resistor measured values

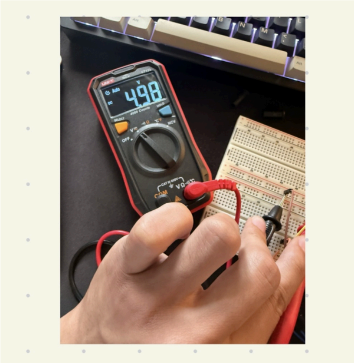


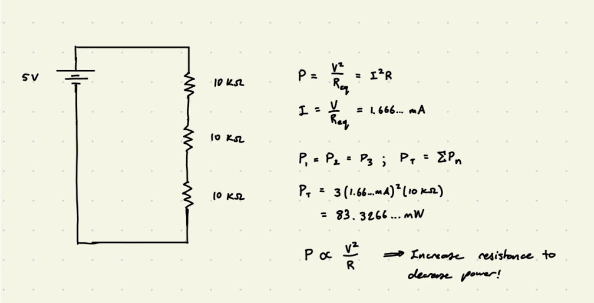
Figure 23,

(Answer red conceptual questions)

The resistance measurement of the resistors needs to be completed outside of the circuit because the voltmeter creates a known current, and then measures the voltage drop accross the resistor. If the resistance is done within the circuit, the power source for the circuit and its existing current will skew the current and make it higher (I assume), resulting in an inaccurate resistance measurement. There is also the complication of currents inducing changes in other currents and what not that exist in non-ideal/real scenarios. As such, to provide a measurement with no external hindrance, the resistor's resistance should be measured outside of the circuit.

(Show circuit analysis and steps to calculate power lost in voltage reduction. Discuss how you could change circuit to reduce this power loss)

```
In [71]: InlineImage('powerin.png', width=13, height=8, caption= "Figure 24, power loss calculation", fs=24);
```



$$P = \frac{V^2}{R_{eq}} = I^2R$$

$$I = \frac{V}{R_{eq}} = 1.666\ldots \text{ mA}$$

$$P_1 = P_2 = P_3 \ ; \ P_T = \Sigma P_n$$

$$P_T = 3(1.66\ldots\text{mA})^2(10\text{ k}\Omega)$$
$$= 83.3266\ldots\text{ mW}$$

$$P \propto \frac{V^2}{R} \Longrightarrow$$ Increase resistance to decrease power!

Figure 24, power loss calculation

(Note any difficulties or troubles and how they were corrected)

No troubles for this section yet. The power lost from the three resistors is approximately 83mW.

(Answer red conceptual questions)

We can see from the equations in figure 24 that power lost is inversely related to the resistance value. Hence, it should be possible to increase the resistance in the circuit to decrease the power lost. As such, in order to reduce the power lost, we should increase the resistance somehow (yet still using 3 'sets' of resistors, so that the voltage is still divided into thirds). A method to do this is to simply add more resistors in series (parallel would reduce resistance), but making sure the amount of resistors is some multiple of 3. This is so we can still access the 3.3v voltage source (2/3 of 5). So: a simple solution is to make a 6 resistor circuit, and the voltage between the last 4 resistors will be approximately 3.3V.

3.2 Task #5: Loaded Voltage Divider

(Indicate with circuit diagram and words how to simulate load and results from measurement)

To simulate a load, I replaced the bridge from the end of the resistors to GND with a 10 kilo-ohm resistor. This likely simulates another device (which typically has an internal resistance) being added to the circuit. The diagram is below, with what the scope reads:

In [75]:

```
InlineImage('groundedresistor.png', width=10, height=8, caption= "Figure 25, load simulation setup", fs=24);
InlineImage('scoped.png', width=10, height=8, caption= "Figure 25, load simulation V reading", fs=24);
```

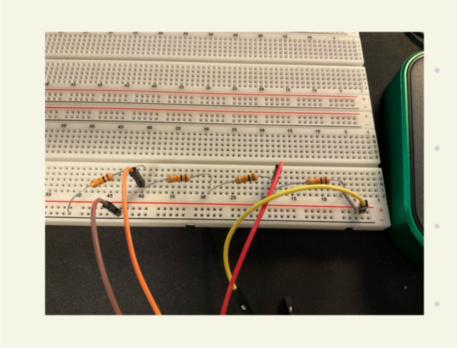
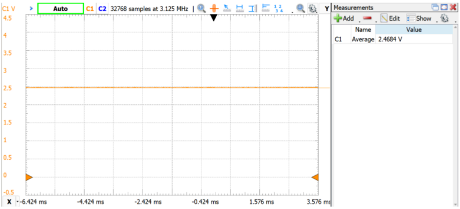


Figure 25, load simulation setup



	Name	Value
C1	Average	2.4684 V

Figure 25, load simulation V reading

(Explain why Vout isn't 3.3 V anymore)

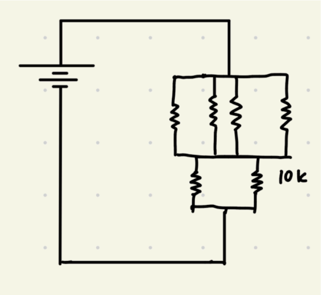
Now that there is another resistor added, the voltage drop is spread out along the fourth one as well; so after each resistor, the voltage drops a quarter of 5 volts. Now, with the oscilloscope probes in the same spots, there is 2 out of 4 resistors between the probes, meaning the voltage drops by half of 5 if measured. (corresponds to the image above). This poses a problem because any subsequent electronic added to this that the 3.3V should power will result in a lowered voltage due to the voltage drop being divided between further circuit components.

(Describe thoughts on how you re-design, diagram it, and build it (with pic). Record results.)

To redesign this circuit, we need something that can guarantee that whatever device attached has access to 3.3V. After thinking for a while, it seems like I just have to assume that we will know the resistance of the device being powered by the 3.3V. Otherwise, I think that it may be impossible to build a circuit given wires and resistors to provide 3.3V with gaurantee to a device with some unknown resistance..? Especially if the device draws a lot of current; the voltage would drop quite fast. In order to use such a voltage divider to power an actual circuit, we may be limited to devices with a very low voltage.

Actually, upon second thought, maybe we can just reduce the resistance so much so that the current in the circuit is super high, and when the load is connected, the current required to run the load will be somewhat negligible in comparison to the circuit, so that the voltage drop accross the resistor stays pretty stable?? This can probably be done by stacking several identical resistors in parallel, so that there are 3 sets of parallel, in series. Or alternatively, it may be more efficient to have 2 sets of parallel, where the first set has double the amount in parallel as the second. (so the first set will have 1/3 of the resistance in the circuit, and the second set will have 2/3) This means that over the second set, there is a drop of 2/3 of the source voltage (which would turn 5v into 3.3v). With this, we can reduce the resistance in the circuit a lot so that a load can use more current without voltage being reduced. HOWEVER: in reality, there is likely an upper limit to how low the resistance can be and how high the current can be... for safety reasons. In this lab, I will just demonstrate this by the following setup:

```
In [77]: InlineImage('revised.png', width=7, height=8, caption= "Figure 26, modified circuit diagram", fs=24);
         InlineImage('revised2.png', width=10, height=8, caption= "Figure 27, modified circuit breadboard", fs=24);
         InlineImage('reviseddata.png', width=114, height=8, caption= "Figure 28, modified circuit scope reading", fs=24);
```



igure 26, modified circuit diagram

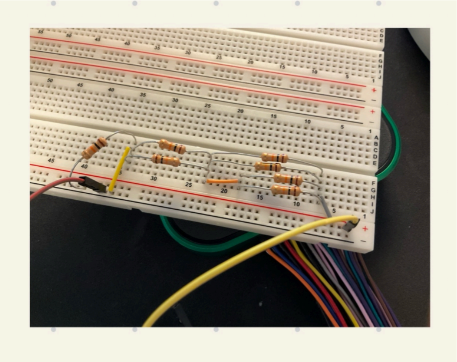
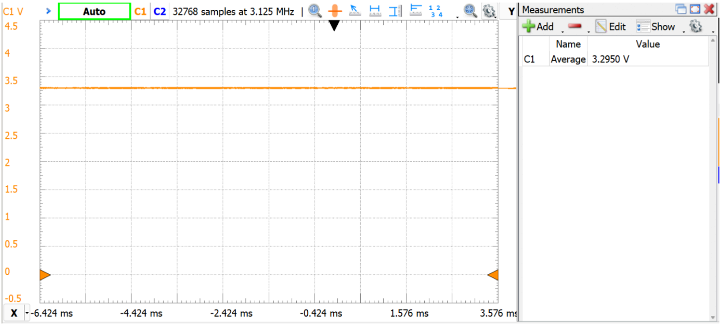


Figure 27, modified circuit breadboard



32768 samples at 3.125 MHz

	Name	Value
C1	Average	3.2950 V

Figure 28, modified circuit scope reading

As we can see, it is a success! With the simulated load, we still have 3.29 volts accross the second set of resistors!

4. Conclusion

(Give a brief summary of the experiment's results, including anything particularly surprising or noteworthy to you.)

This lab clearly demonstrated some basics of using the oscilloscope. In particular, finding the right scope settings to see the wave is very important. I am now familiar with importing data and graphing it here in jupyter. I also recognize the need to distinguish between the positive and negative probes when taking measurements with the oscilloscope/AD3. With this lab, basic circuit behaviour was demonstrated, and we were able to analyze how a theoretical drawing of a circuit behaves in the real world.

For the unloaded circuit, there were clear differences between the expected values for the voltage and what we actually measured. This is due to power supplies and resistors in reality not being exactly what they say; there is always some degree of error. My laptop and the AD3 are not a perfect power supply either; according to the waveforms software, my laptop only outputs about 4.825 volts through the usb, when the power is set to 5V. I noticed that 5V also seems to be the maximum voltage my laptop can output through the USB. Howver, there is another power supply for the AD3 that plugs into the wall- I have not tested that, but I believe there is potential for a voltage supply higher than 5V.

For the loaded circuit, I had a lot of difficulty coming up with an idea that would work; I initially thought that it would be impossible to build such a circuit without knowing the internal resistance of the device to be powered by the 3.3V. Because how else do you provide a 3.3 volt voltage drop? My first idea was to build a circuit where there is a voltage drop of 3.3V somewhere, and then plug in the device in parallel with the 3.3V voltage drop. However, I realized that this is essentially like measuring the unloaded circuit. As soon as I add something eg a resistor, the resistance in that branch changes and the voltage drop is no longer 3.3V. It seems impossible, so I decided to move forward assuming the resistance of the device/load would simply be 10k, equivalent to the 10k resistor we add in 3.2.2 to simulate a load. However, I eventually realized that we don't have to restructure the circuit too much, rather we should aim to increase the current so that any current drawn by a load will be miniscule in comparison to that in the circuit, so that V = IR still holds stable (since 'I' will change less). All in all, it took a bit of thinking but this lab has demonstrated a lot about the basics of circuitry.

In [1]:

```python
# comments in Python start with a "#" and will not be executed - well commented code is easier for graders to understand and give partial marks to ;)
# Importing modules that will be necessary for the tutorial
import numpy as np # numpy is a general math/array manipulation library that is almost always used in any scientific/engineering notebook
import matplotlib.image as mplimg # matplotlib.image is used for displaying images in python
import matplotlib.pyplot as plt # matplotlib.pyplot is used for general plotting and is very similar to matlab syntax for plotting
import pandas as pd # pandas is a data manipulation library, we will use it to import data from excel files
print('All necessary modules imported!')
```

```
All necessary modules imported!
```

In [2]:

```python
def InlineImage(img, width=8, height=8, caption=None, fs=24):
    # Author: BG
    # Renders an image with a caption underneath using matplotlib
    # Requires matplotlib.pyplot to be imported as plt and matplotlib.img to be imported as mplimg
    # img - string, the location of an image file
    # width, height - numbers, the dimensions of the figure the image is displayed in. Original aspect ratio of the image will be preserved
    # caption - string, the caption to be displayed beneath the image
    # fs - number, the font size of the caption
    fig,ax = plt.subplots(figsize=(width,height))
    ax.imshow(mplimg.imread(img))
    ax.set_xlabel(caption,fontsize=fs)
    ax.tick_params(
        which='both',
        bottom=False,
        top=False,
        left=False,
        labelleft=False,
        labelbottom=False)
    ax.spines["top"].set_visible(False)
    ax.spines["left"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.spines["bottom"].set_visible(False)
```

In [34]:

```python
import os

# List all PNG files in the current directory
png_files = [f for f in os.listdir() if f.lower().endswith(".png")]
print("Current directory:", os.getcwd())
print("PNG files found:", png_files)
```

```
Current directory: /home/jupyter/ENPH259
PNG files found: []
```

In [37]:

```python
import os
os.getcwd()

import os

# List all PNG files in the current directory
png_files = [f for f in os.listdir() if f.lower().endswith(".png")]
print("Current directory:", os.getcwd())
print("PNG files found:", png_files)
```

```
Current directory: /home/jupyter/ENPH259
PNG files found: []
```